

Worksheet4

Changze Wu_1008837840

2024-02-16

```
qnorm(c(0.25, 0.75), mean = 50, sd = 3)
```

```
## [1] 47.97653 52.02347
```

```
qnorm(0.9, mean = 50, sd =3)
```

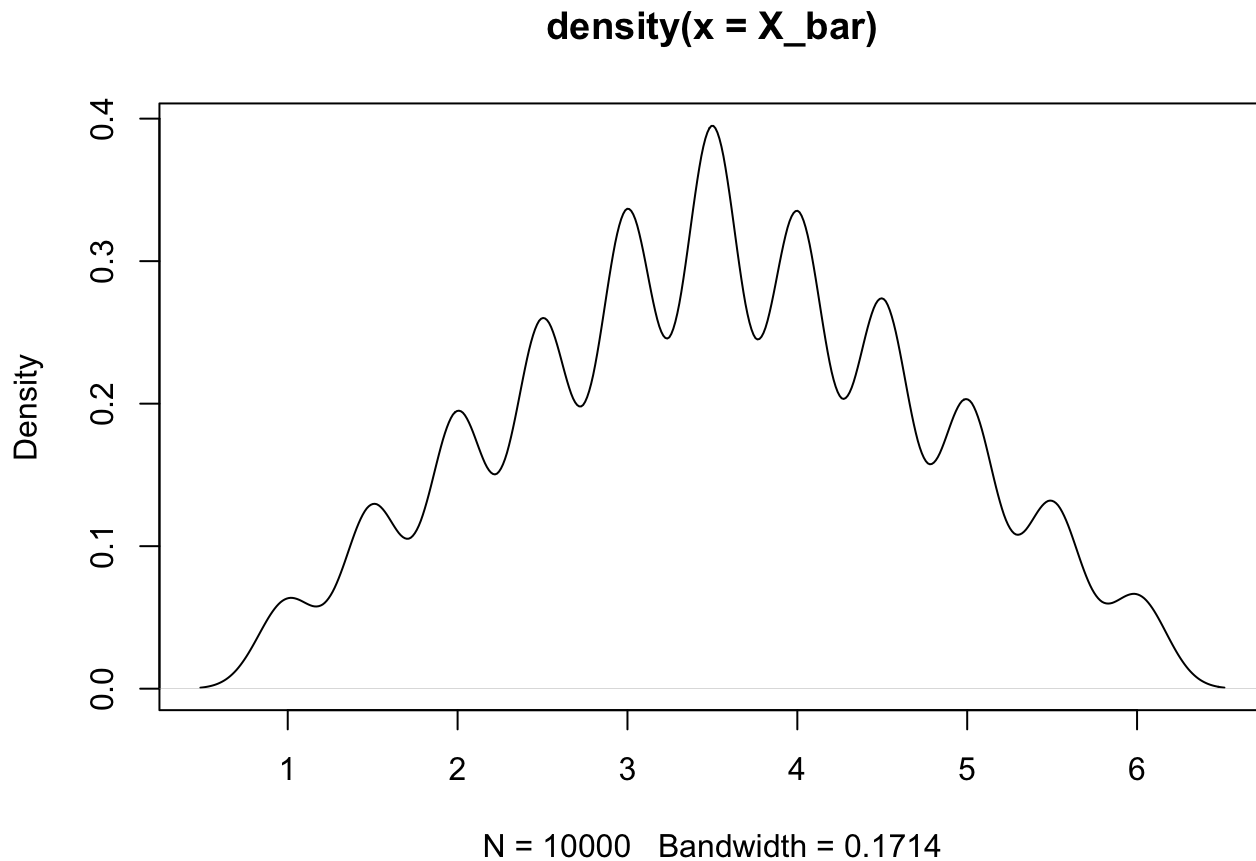
```
## [1] 53.84465
```

```
qnorm(c(0.1, 0.9), mean = 50, sd =3)
```

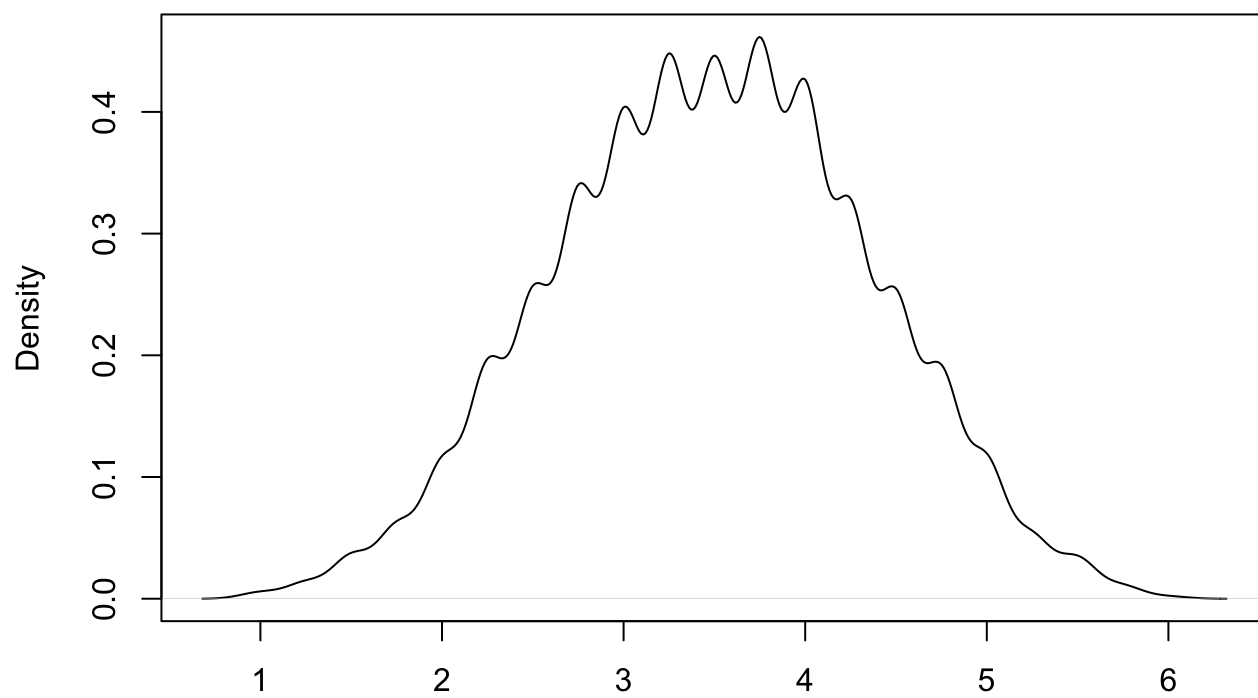
```
## [1] 46.15535 53.84465
```

```
sample_die = function(n){  
  s = sample(c(1, 2, 3, 4, 5, 6), size = n, replace = T)  
  return(mean(s))  
}
```

```
X_bar = replicate(10000, sample_die(n = 2))  
plot(density(X_bar))
```

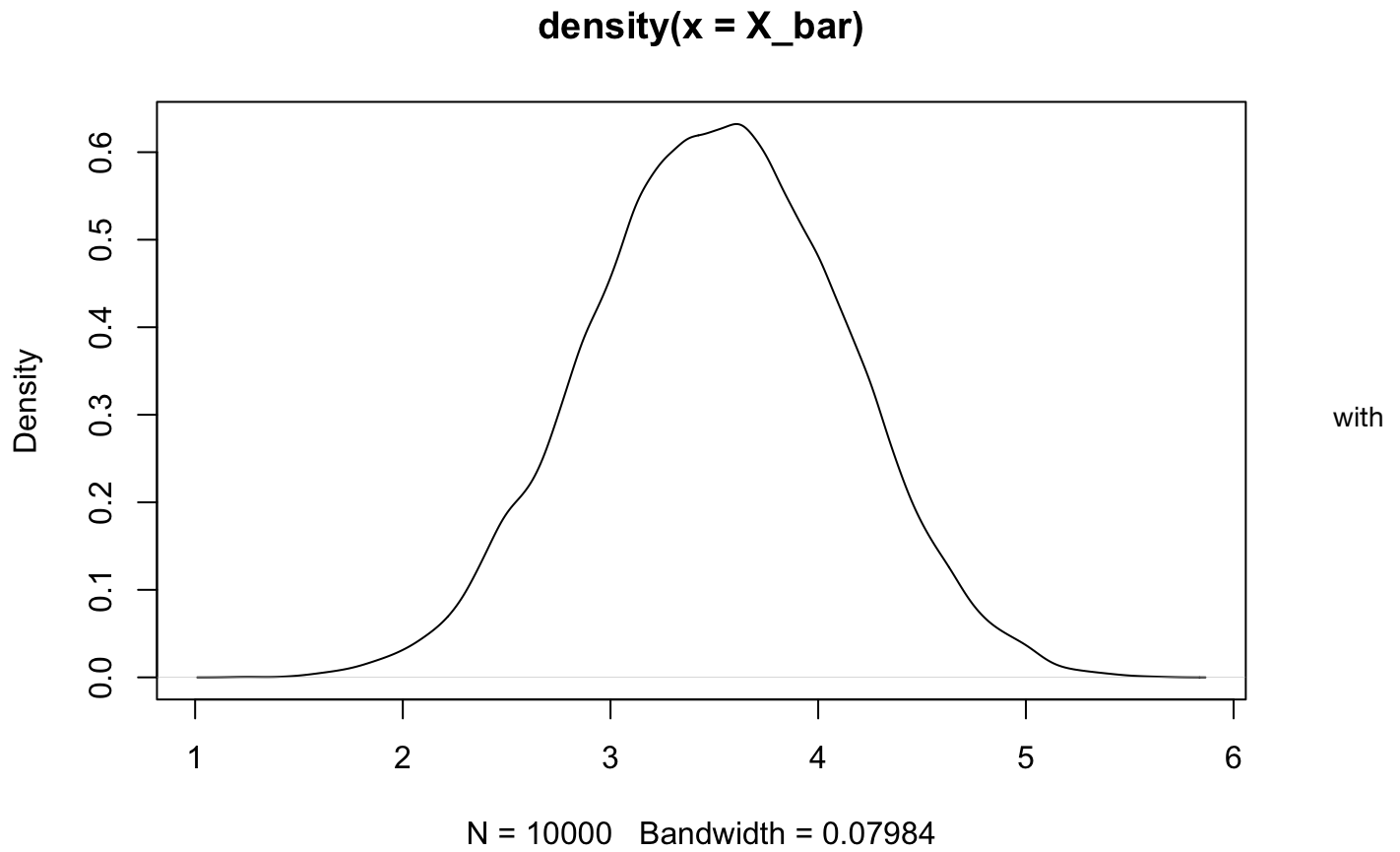


```
X_bar = replicate(10000, sample_die(n = 4))  
plot(density(X_bar))
```

density(x = X_bar)

N = 10000 Bandwidth = 0.1064

```
X_bar = replicate(10000, sample_die(n = 8))  
plot(density(X_bar))
```



the increasing value of n , the graph becomes more likely to normal distribution. if the n is big enough, the graph becomes normal distribution